Practical task 5: Covid-19-dataset

Import libraries and load dataset:

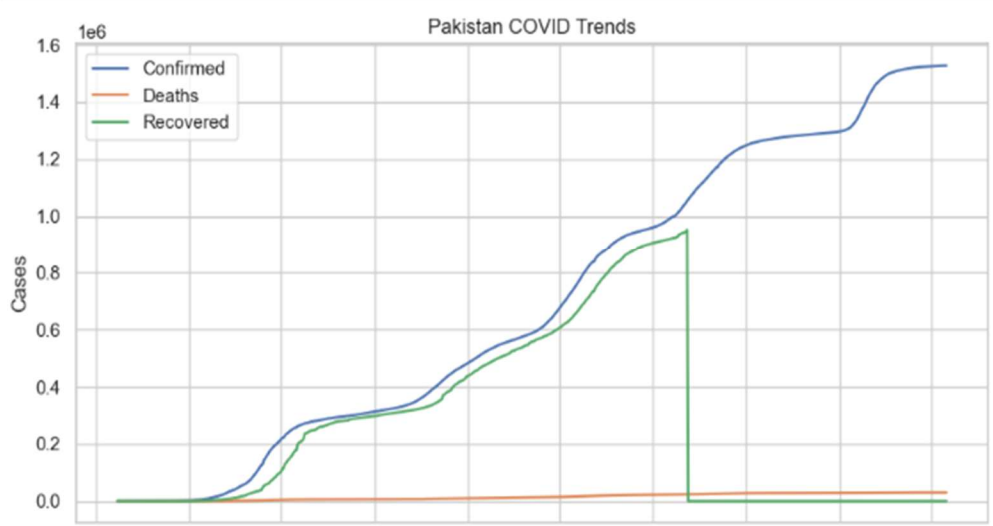
```
•[2]: #Import libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

sns.set(style="whitegrid")
```

```
•[17]: # load dataset
url = "https://raw.githubusercontent.com/datasets/covid-19/main/data/countries-aggregated.csv"
df = pd.read_csv(url)

df['Date'] = pd.to_datetime(df['Date'])

# filter Pakistan data
country_df = df[df['Country'] == 'Pakistan']

print(country_df.head())
```

```
             Date   Country  Confirmed  Recovered  Deaths
109344 2020-01-22  Pakistan          0          0       0
109345 2020-01-23  Pakistan          0          0       0
109346 2020-01-24  Pakistan          0          0       0
109347 2020-01-25  Pakistan          0          0       0
109348 2020-01-26  Pakistan          0          0       0
```

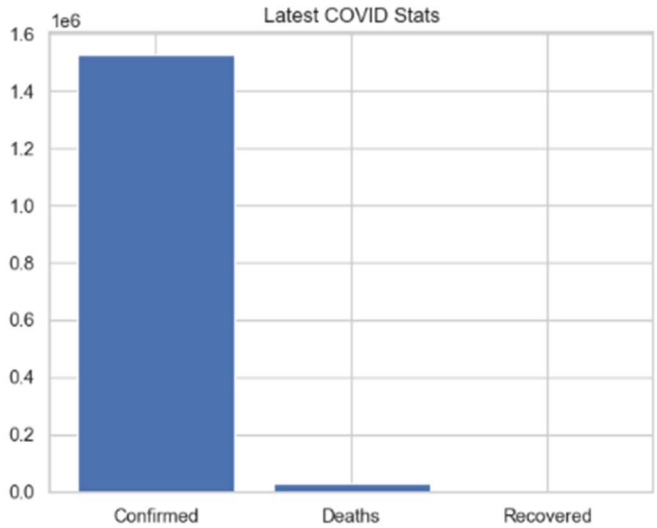
Line plot:

```
•[4]: #line plot
plt.figure(figsize=(10,5))

plt.plot(country_df['Date'], country_df['Confirmed'], label='Confirmed')
plt.plot(country_df['Date'], country_df['Deaths'], label='Deaths')
plt.plot(country_df['Date'], country_df['Recovered'], label='Recovered')

plt.title("Pakistan COVID Trends")
plt.xlabel("Date")
plt.ylabel("Cases")
plt.legend()

plt.show()
```



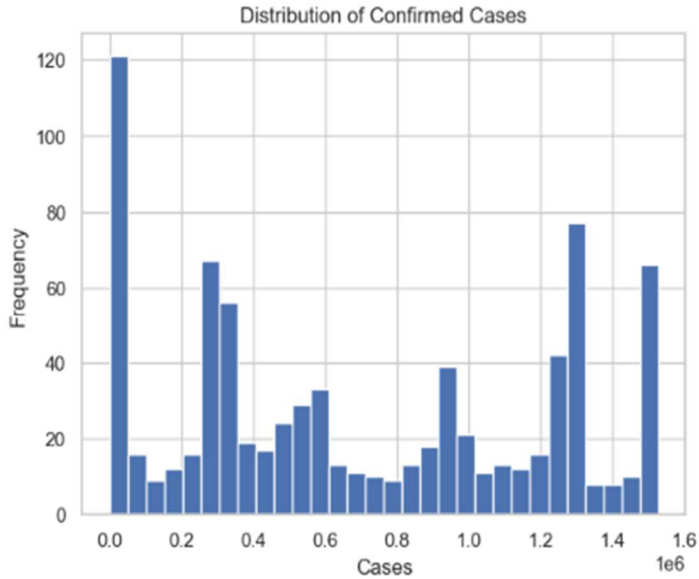
Bar chart:

```
[5]: #Bar chart
latest = country_df.tail(1)

plt.bar(['Confirmed', 'Deaths', 'Recovered'],
        [latest['Confirmed'].values[0],
         latest['Deaths'].values[0],
         latest['Recovered'].values[0]])

plt.title("Latest COVID Stats")
plt.show()
```



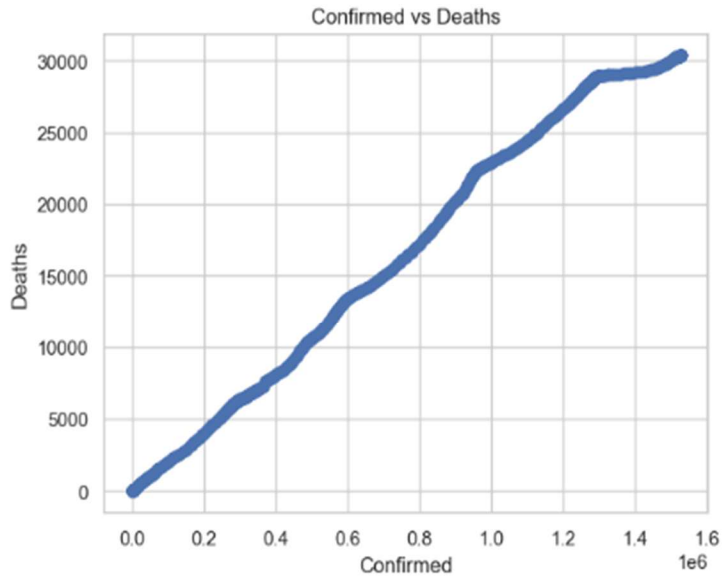
Histogram:

```
[6]: #Histogram
plt.hist(country_df['Confirmed'], bins=30)

plt.title("Distribution of Confirmed Cases")
plt.xlabel("Cases")
plt.ylabel("Frequency")

plt.show()
```



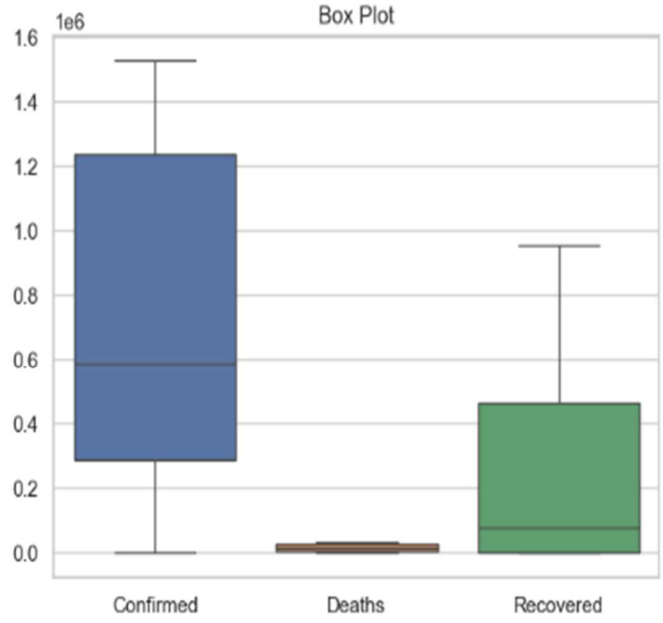
Scatter plot:

```
[7]: #Scatter plot
plt.scatter(country_df['Confirmed'], country_df['Deaths'])

plt.xlabel("Confirmed")
plt.ylabel("Deaths")
plt.title("Confirmed vs Deaths")

plt.show()
```



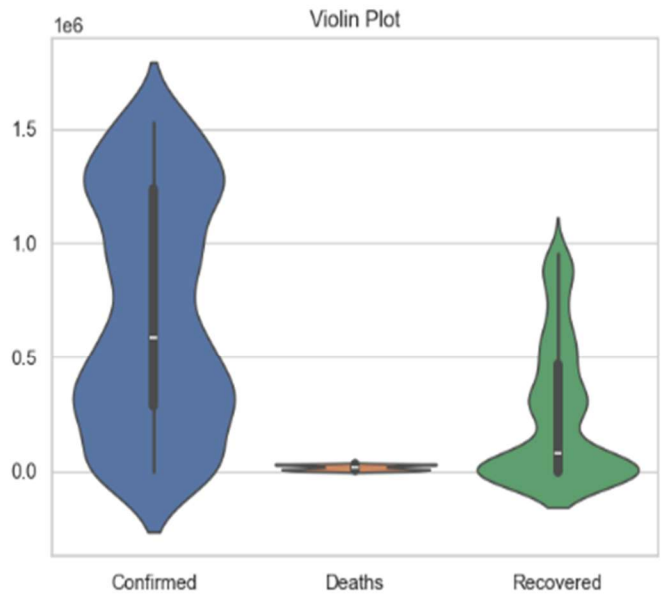
Box plot:

```
[8]: #box plot
sns.boxplot(data=country_df[['Confirmed','Deaths','Recovered']])

plt.title("Box Plot")
plt.show()
```



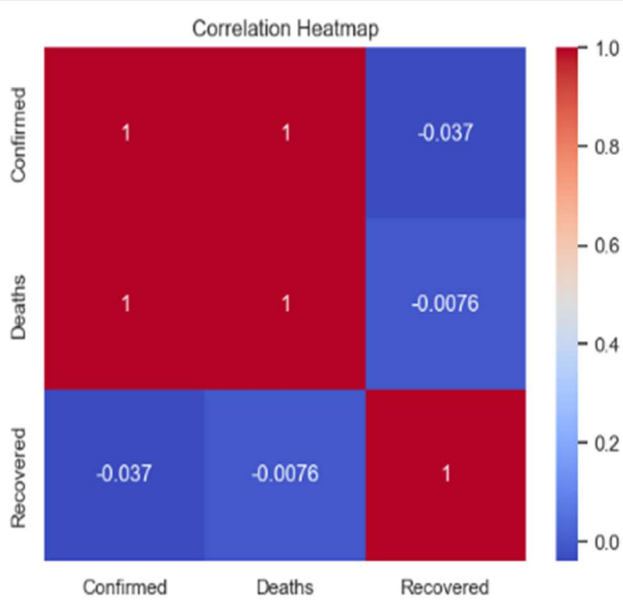
Violin plot:

```
[9]: #violin plot
sns.violinplot(data=country_df[['Confirmed','Deaths','Recovered']])

plt.title("Violin Plot")
plt.show()
```



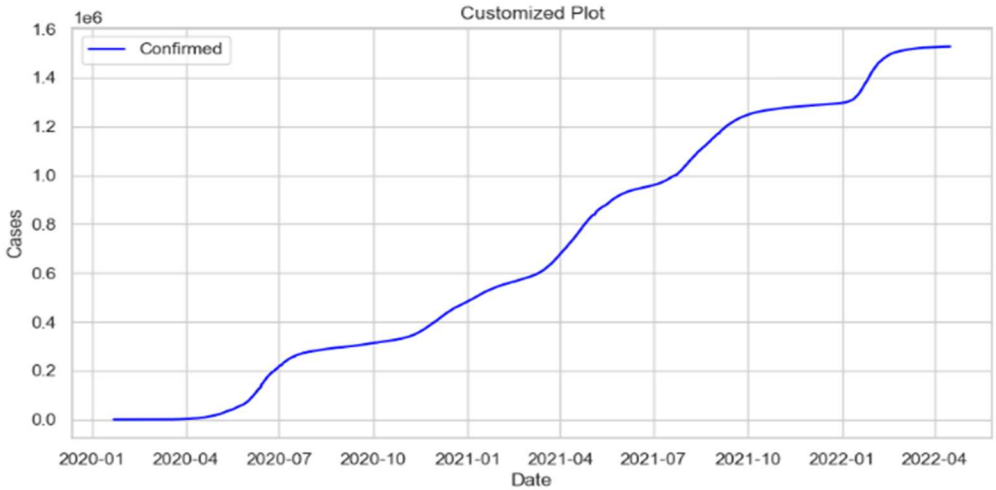
Heatmap:

```
[10]: #Heatmap
corr = country_df[['Confirmed','Deaths','Recovered']].corr()

sns.heatmap(corr, annot=True, cmap='coolwarm')

plt.title("Correlation Heatmap")
plt.show()
```



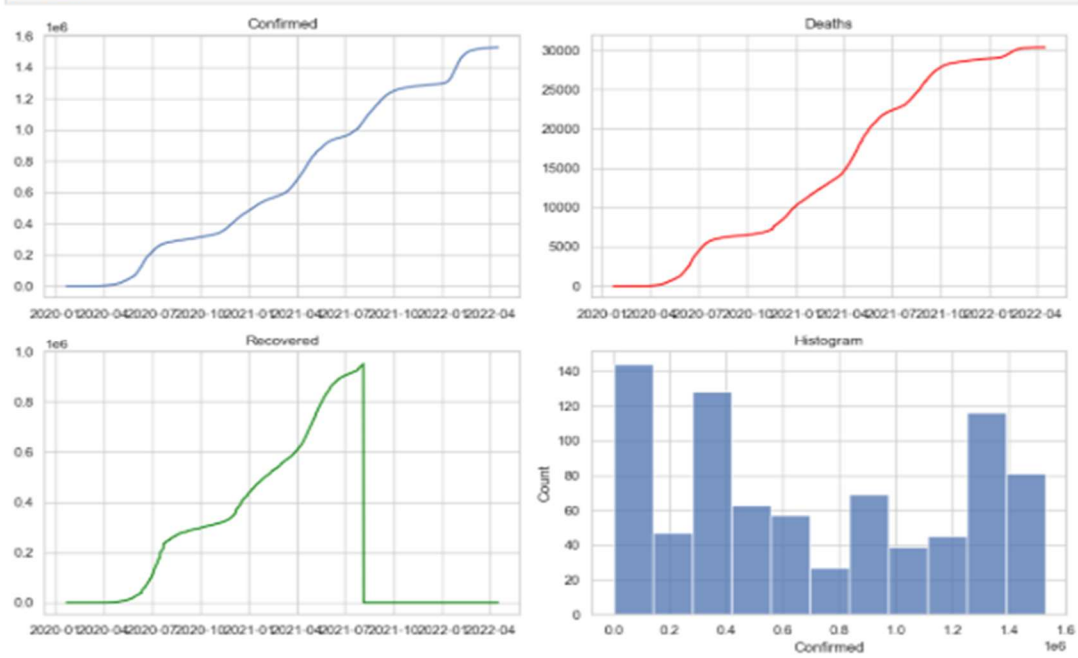
Customize plot:

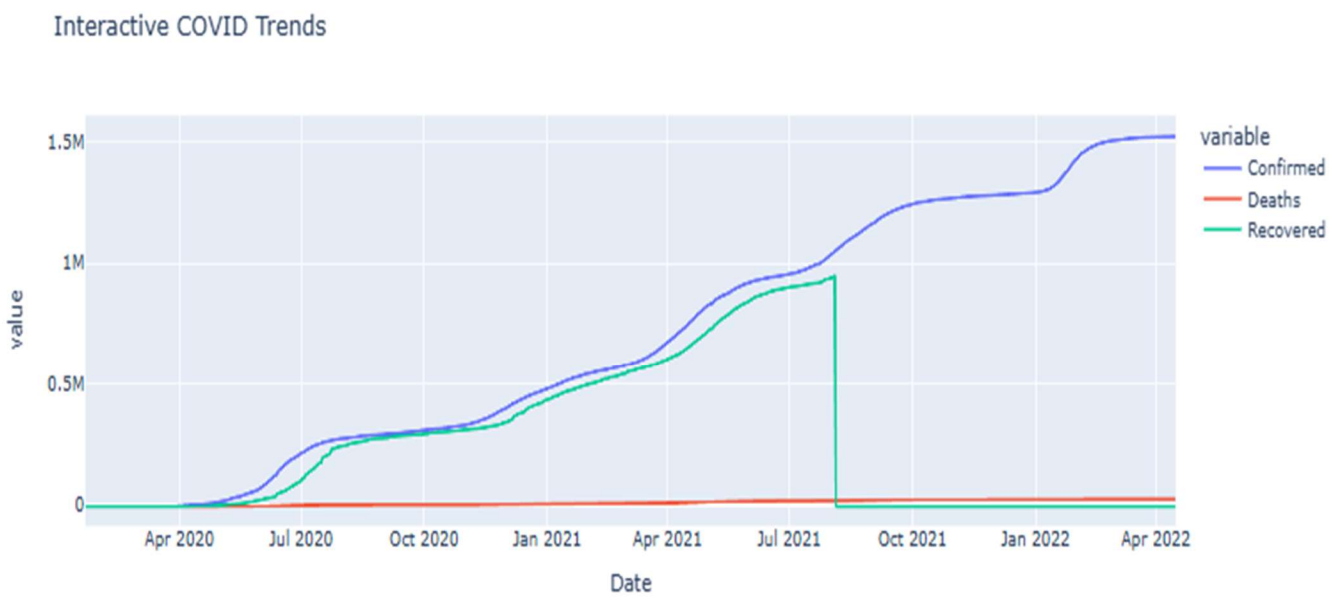
```
*[11]: #Customize Plots
plt.figure(figsize=(10,5))

plt.plot(country_df['Date'], country_df['Confirmed'], color='blue', label='Confirmed')

plt.title("Customized Plot")
plt.xlabel("Date")
plt.ylabel("Cases")
plt.legend()
plt.grid(True)

plt.show()
```



Subplot/multi-plane figure:

```
*[12]: #Subplots / Multi-panel Figures
fig, axs = plt.subplots(2, 2, figsize=(12,8))

axs[0,0].plot(country_df['Date'], country_df['Confirmed'])
axs[0,0].set_title("Confirmed")

axs[0,1].plot(country_df['Date'], country_df['Deaths'], color='red')
axs[0,1].set_title("Deaths")

axs[1,0].plot(country_df['Date'], country_df['Recovered'], color='green')
axs[1,0].set_title("Recovered")

sns.histplot(country_df['Confirmed'], ax=axs[1,1])
axs[1,1].set_title("Histogram")

plt.tight_layout()
plt.show()
```



Iteractive plot with plotly:

•[13]:
```
#Interactive Plot with Plotly
fig = px.line(country_df, x='Date', y=['Confirmed','Deaths','Recovered'],
              title="Interactive COVID Trends")

fig.show()
```